

# Metro Ticket Booking System

Proposed Solution Architecture (Project Requirement & Design Phase)

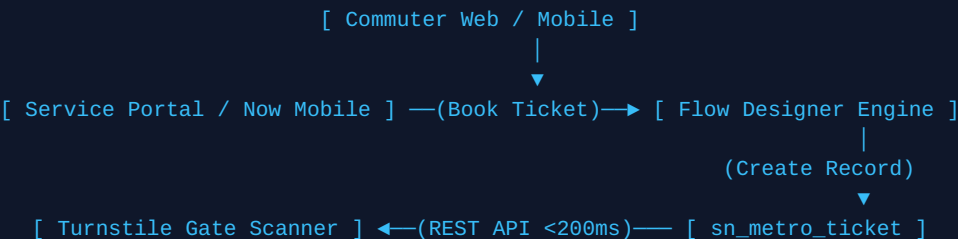
TARGET PLATFORM  
ServiceNow App Engine

DOCUMENT TYPE  
Proposed Solution Specification

APPLICATION SCOPE  
x\_snc\_metro\_book

## 1. High-Level Architecture Overview

The proposed solution leverages ServiceNow **App Engine**, **Service Portal**, **Integration Hub**, and **Scripted REST APIs** to create an end-to-end ticketing platform bridging commuter web/mobile access with physical turnstile gate hardware.



## 2. End-to-End Proposed Solution Modules

### Module A: Commuter Booking & Pass Purchasing (Front-End)

- **Service Portal Catalog Item:** Commuters select origin station, destination station, journey type (Single, Return, Weekly Pass), and passenger category.
- **Real-Time Fare Estimator:** Service Portal Client Script calls a Script Include that evaluates the fare matrix table (sn\_metro\_fare\_matrix) to display live pricing prior to checkout.
- **Mobile Pass Display:** Active tickets are accessible via the **Now Mobile App**, rendering dynamic QR codes through a custom Canvas Widget.

### Module B: Turnstile Gate Inbound Integration (<200ms Validation)

- **Inbound Scripted REST API:** Hardware turnstiles issue POST requests to /api/x\_metro/gate\_validation/scan containing station ID and QR token hash.
- **Fast-Path Verification:** Endpoint performs indexed queries against sn\_metro\_ticket using composite primary keys and hashes.
- **Response Payload:** Returns immediate decisions: {"status": "ALLOW", "gate\_action": "OPEN"} or {"status": "DENY", "reason": "INVALID\_TOKEN"}.
- **Async Logging:** Ticket state updates (e.g., Active to In-Transit) execute asynchronously via Business Rules to prevent gate delay.

### Module C: Payment Gateway & Automated Exception Handling

- **Outbound Payment Spokes:** Integration Hub connects with external payment gateways (Stripe, Razorpay, UPI) to initiate secure transactions.

- **Idempotent Webhook Listener:** Listens for payment gateway callbacks and validates transaction IDs to prevent duplicate ticket generation.
- **Auto-Refund Flow:** If payment succeeds but ticket activation fails, a 60-second Flow Designer timer triggers an automated refund step via Integration Hub.

Module D: Automated Expiration & Housekeeping

- **Scheduled Script Executions:** Background jobs run every 15 minutes to evaluate active single-journey tickets. Any unused ticket exceeding its validity window (e.g., 3 hours post-purchase) transitions automatically to Expired.

3. Core Database Schema & Relationships

| Table Name           | Extends    | Core Fields                                                         | Purpose                         |
|----------------------|------------|---------------------------------------------------------------------|---------------------------------|
| sn_metro_station     | Base Table | name, code, line_color, active                                      | Station master record.          |
| sn_metro_fare_matrix | Base Table | origin_station, dest_station, base_fare, peak_multiplier            | Zone & distance pricing lookup. |
| sn_metro_ticket      | task       | number, commuter, origin, destination, state, qr_token, valid_until | Core ticket record lifecycle.   |
| sn_metro_payment     | Base Table | ticket, transaction_id, amount, payment_status, gateway_response    | Financial audit ledger.         |

4. Key ServiceNow Deliverables (Artifact Mapping)

| Requirement         | Proposed ServiceNow Artifact                 | Design Benefit                                                       |
|---------------------|----------------------------------------------|----------------------------------------------------------------------|
| Commuter Interface  | Service Portal Custom Widget & Now Mobile    | Mobile-friendly purchase and responsive QR rendering.                |
| Gate Scanners       | Scripted REST API (/api/x_metro/...)         | Sub-200ms latency validation for turnstile hardware.                 |
| Fare Logic          | Decision Tables Engine & Script Include      | Configurable dynamic pricing without hardcoded logic.                |
| Payment Integration | Integration Hub Outbound REST Spoke          | Secure third-party payment processing & webhooks.                    |
| Ticket Lifecycle    | Business Rules & Scheduled Script Executions | Automated state management (Active → In-Transit → Closed / Expired). |